5 Set up the eggs

Now make some variables to store the colours, width, and height of the eggs. You'll also need variables for the score, the speed of the falling eggs, and the interval between new eggs appearing on the screen. The amount they are changed by is determined by the `difficulty_factor` – a lower value for this variable actually makes the game harder.

```
c.pack()

color_cycle = cycle(['light blue', 'light green', 'light pink', 'light yellow', 'light cyan'])
egg_width = 45
egg_height = 55
egg_score = 10
egg_speed = 500
egg_interval = 4000
difficulty_factor = 0.95
```

The `cycle()` function allows you to use each colour in turn.

You score 10 points for catching an egg.

A new egg appears every 4,000 milliseconds (4 seconds).

This is how much the speed and interval change after each catch (closer to 1 is easier).

6 Set up the catcher

Next add the variables for the catcher. As well as variables for its colour and size, there are four variables that store the catcher's starting position. The values for these are calculated using the sizes of the canvas and the catcher. Once these have been calculated, they are used to create the arc that the game uses for the catcher.

Don't forget to save your work.

```
difficulty_factor = 0.95

catcher_color = 'blue'
catcher_width = 100
catcher_height = 100
catcher_start_x = canvas_width / 2 - catcher_width / 2
catcher_start_y = canvas_height - catcher_height - 20
catcher_start_x2 = catcher_start_x + catcher_width
catcher_start_y2 = catcher_start_y + catcher_height

catcher = c.create_arc(catcher_start_x, catcher_start_y, \
                       catcher_start_x2, catcher_start_y2, start=200, extent=140, \
                       style='arc', outline=catcher_color, width=3)
```

This is the height of the circle that is used to draw the arc.

These lines make the catcher start near the bottom of the canvas, in the centre of the window.

Start drawing at 200 degrees on the circle.

Draw for 140 degrees.

Draw the catcher.